

A Network-Based Approach to Identify the Toxicity Mechanisms of Silver Nanoparticles and Chlorpyrifos in *Danio rerio* (Zebrafish)

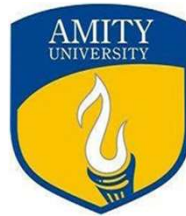
Presented by

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INTRODUCTION

❖ **Network Biology:** It studies biological systems by mapping interactions between genes, proteins, and metabolites.

Application: We can identify which proteins interact in zebrafish when exposed to AgNPs and chlorpyrifos, predicting disrupted pathways.

❖ **Protein-Protein Interactions (PPIs):** Proteins rarely act alone; they form complex networks to perform biological functions.

Application: To map how AgNPs and chlorpyrifos alter PPI networks in zebrafish.

❖ **Silver Nanoparticles (AgNPs) and Chlorpyrifos (CPF):**

AgNPs: Widely used in medicine, cosmetics, and industry, but concerns about nanotoxicity in agriculture.

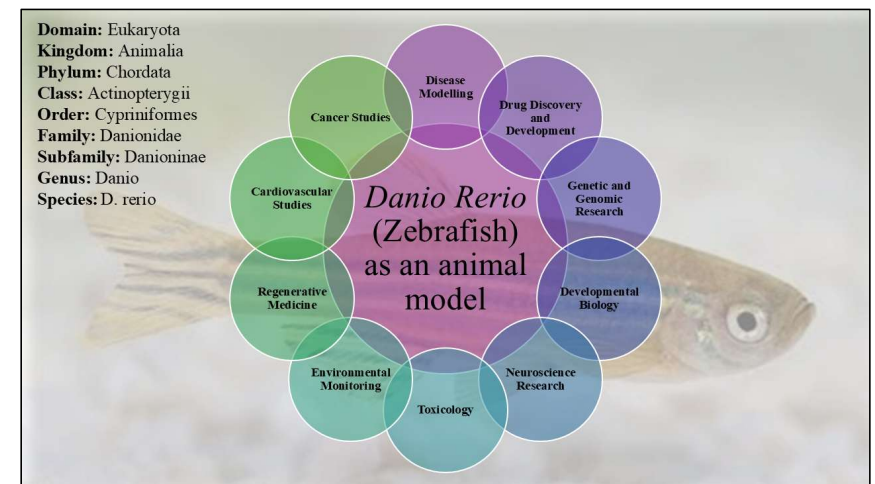
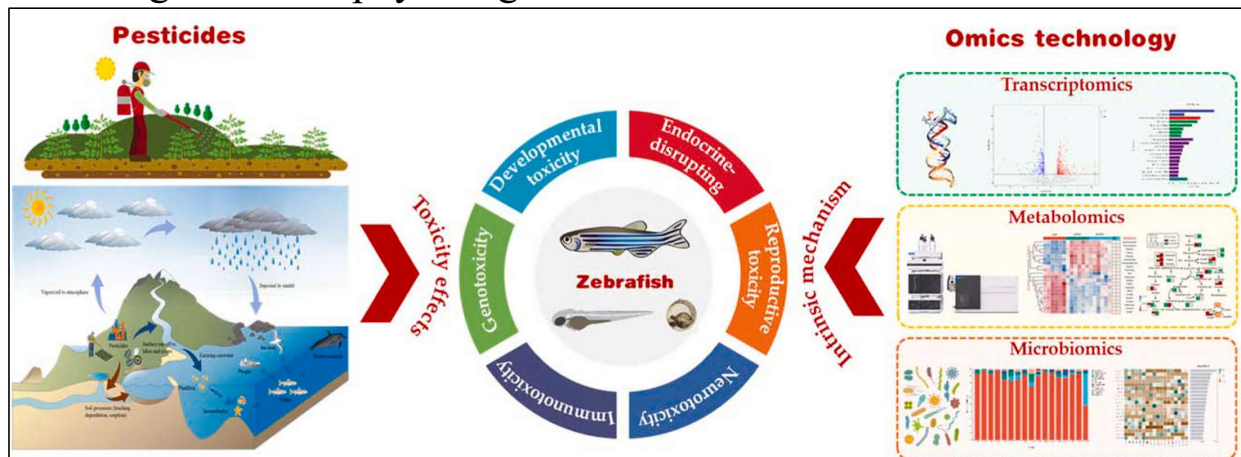
CPF: A common pesticide with known neurotoxic and endocrine-disrupting effects.

Both chemicals enter aquatic environments, affecting organisms like zebrafish.

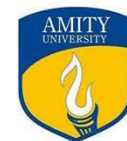
❖ **Zebrafish (*Danio rerio*):**

A widely used model organism in toxicology studies, fast reproduction and transparent embryos allow easy observation.

Shares genetic and physiological similarities with humans.



REVIEW OF LITERATURE



- Chlorpyrifos (CPF) is a widely studied organophosphate pesticide with documented neurodevelopmental toxicity, especially in aquatic organisms. Studies have shown that CPF inhibits acetylcholinesterase (AChE), leading to impaired neurotransmission and developmental anomalies in zebrafish embryos (Slotkin, 2004; Richendrfer et al., 2012). It also causes oxidative stress and apoptosis in neural and hepatic tissues (Song et al., 2012). Silver nanoparticles (AgNPs), increasingly prevalent in commercial and medical applications, have also raised environmental concerns due to their bioaccumulation and toxic effects in aquatic ecosystems. Research indicates that AgNPs induce developmental malformations, oxidative stress, and immunotoxicity in zebrafish and other aquatic species (Griffitt et al., 2008; Bar-Ilan et al., 2009).
- Zebrafish (*Danio rerio*) has emerged as a robust model for toxicological studies due to its genetic similarity to humans, rapid development, and transparency of embryos (Howe et al., 2013). Its utility in high-throughput screening and systems-level toxicology makes it ideal for investigating long-term effects of environmental pollutants such as CPF and AgNPs (Hill et al., 2005; Reimers et al., 2006). Notably, several studies have established the zebrafish brain as a sensitive organ for evaluating neurotoxicity using molecular and behavioral endpoints (Ali et al., 2011; Kalueff et al., 2014).
- Network biology has proven to be a powerful tool in systems toxicology. By analyzing protein-protein interaction (PPI) networks, researchers can uncover key hub proteins and regulatory modules perturbed by toxicants (Barabási et al., 2011; Shannon et al., 2003). Tools like STRING and Cytoscape have facilitated the visualization and analysis of such networks, offering insights into mechanistic pathways underlying toxicity. Integration of network modules with enrichment analysis enables the identification of affected biological processes and pathways, which may not be apparent through single-gene studies.

- Single-cell RNA sequencing (scRNA-seq) provides cell-type-specific expression data that is particularly useful in neurotoxicology. This technique allows detection of subtle transcriptional changes across individual brain cells, making it invaluable for assessing the impact of CPF and AgNPs on neural circuitry (Wagner et al., 2016; Farrell et al., 2018). Recent zebrafish scRNA-seq datasets, such as GSE134288, have enabled such high-resolution investigations.
- Molecular docking is another well-established computational approach used to assess chemical-protein interactions. It predicts binding affinities and interaction sites of small molecules with target proteins, helping to hypothesize potential disruption of biological functions. CPF has been shown to interact with enzymes like CYP450s and SOD1, contributing to its toxicodynamic effects (Choudhary et al., 2011; Muthukumaran et al., 2013). However, limited structural data for AgNPs has hindered similar analysis, although nano-informatics approaches are being developed to overcome this.
- Machine learning (ML) techniques are increasingly applied in predictive toxicology for modeling complex relationships between chemical descriptors and toxicity outcomes. Earlier models largely focused on quantitative structure-activity relationships (QSARs), but newer approaches now integrate biological features such as pathway enrichment data to improve interpretability and performance (Mayr et al., 2016; Huang et al., 2017). Although advanced ML platforms are available, even simplified models, such as those implemented in Excel, can demonstrate valuable classification accuracy when informed by mechanistic features.
- The growing trend in toxicology is toward integrated, multi-omics approaches that combine transcriptomics, proteomics, docking, and ML into cohesive frameworks. Such systems toxicology models enhance predictive power, mechanistic clarity, and ethical acceptability by reducing animal testing (Hartung, 2009; Krewski et al., 2020). These approaches are particularly crucial for assessing complex exposures, such as those posed by mixtures or emerging materials like nanoparticles, where traditional toxicological testing falls short.

OBJECTIVES

1. To identify key regulatory proteins and biological pathways disrupted by silver nanoparticles and chlorpyrifos in zebrafish using a network biology approach.
2. To integrate RNA-seq transcriptomic data with protein interaction networks for toxicity analysis.
3. To perform molecular docking studies to predict the interaction of silver nanoparticles and chlorpyrifos with zebrafish metabolic enzymes.
4. To develop a machine learning-based model for toxicity scoring based on long-term risk assessment.

RATIONALE AND HYPOTHESIS



Rationale:

Traditional toxicology studies often overlooks the interconnected nature of biological responses and may fail to detect subtle but biologically significant effects of sub-lethal exposures. By integrating molecular docking, transcriptomics, pathway enrichment, and network biology, we can capture the complex cascade of molecular events triggered by toxicants like Chlorpyrifos (CPF) and Silver Nanoparticles (AgNPs) in zebrafish.

Hypothesis:

CPF and AgNP exert long-term toxicity in zebrafish by disrupting key stress response, neurodevelopmental, and metabolic networks, leading to sustained biological dysfunction. These effects can be predicted through computational models, thereby providing a functionally integrated toxicity score.

METHODOLOGY

1.1. Data collection

Extracted zebrafish-specific genes associated with chlorpyrifos (CPF) and silver nanoparticles (AgNPs)

Tools used: Comaparative Toxicogenomics Database (CTDbase)

1.2. Cleaning the Raw data

Removed duplicates and extra interacting chemicals. Filtered data with reference to literature and experimental data.

Extract data into key columns: Chemical Name, Chemical ID, Gene symbol, Interaction type, and expression.

1.3. Construction of PPI Network

Imported gene list to STRING (v11.5) database with high confidence score ≥ 0.7 .

To visualise the network file was exported to Cytoscape software. Clusters of network were identified.

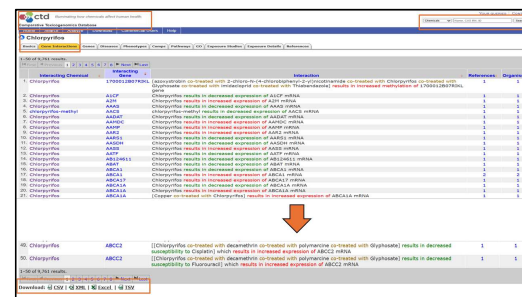
1.4. Visualization and Analysis in Cytoscape

Hub proteins were identified using the CytoHubba plugin, based on centrality metrics such as Degree, Betweenness, and MCC (Maximal Clique Centrality).

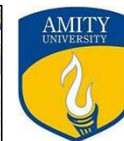
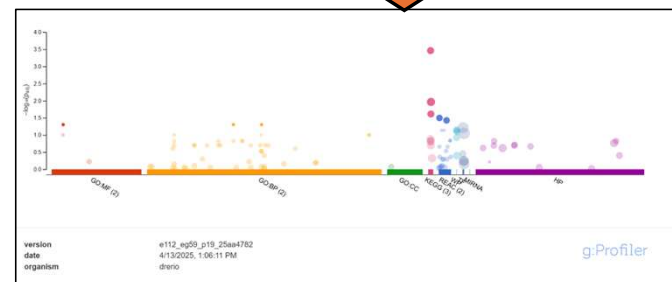
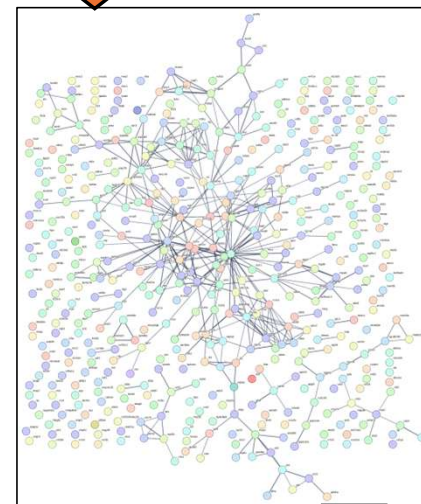
1.5. Functional Enrichment Analysis

The g:Profiler tool was employed, the number of significantly affected pathways were tabulated and later used as a toxicity feature.

Highlighted pathways relevant to stress response, apoptosis, oxidative damage, and neurodevelopment.



Chemical Name	Chemical ID	Gene Symbol	Interaction Type	Expression
Chlorpyrifos	1700012807000	ABCC2	Chlorpyrifos co-treated with desferrioxime co-treated with polyethylene co-treated with dihydroxyacetone results in increased expression of ABCC2 mRNA	1
Chlorpyrifos	ABCC2	ABCC2	Chlorpyrifos co-treated with desferrioxime co-treated with polyethylene co-treated with dihydroxyacetone results in increased expression of ABCC2 mRNA	1
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Chlorpyrifos	ABCC2	ABCC2	Chlorpyrifos co-treated with desferrioxime co-treated with polyethylene co-treated with dihydroxyacetone results in increased expression of ABCC2 mRNA	1



2.1. Data selection

Retrieved zebrafish brain single cell RNA-seq dataset (GSE134288, 10X Genomics) from NCBI GEO.

This dataset contained ~6,500 zebrafish brain cells collected from transgenic *nr1d1:VNP* larvae at 6.5 days post-fertilization

2.2. Data Preprocessing

The gene-cell expression matrix was loaded and pre-processed using Python (v3.10) and the Scanpy library.

Quality control filters were applied to retain cells with >500 genes expressed, and genes detected in >3 cells.

Mitochondrial genes and low-abundance transcripts were removed.

```
C:\Users> sara > OneDrive > Desktop > filtered_scRNAseq_matrix.csv
1 import pandas as pd
2 import numpy as np
3 import scanpy as sc
4 import matplotlib.pyplot as plt
5 import seaborn as sns
6
7 expression_data = pd.read_csv("GSM941431_gene_cell_matrix.txt", sep="\t", index_col=0)
8
9 print("Matrix shape:", expression_data.shape) # Genes x Cells
10 print("First 5 rows of the data:")
11 print(expression_data.head())
12
13 min_genes_per_cell = 500
14 cell_sums = (expression_data > 0).sum(axis=0)
15 filtered_cells = cell_sums[cell_sums > min_genes_per_cell].index
16 filtered_data = expression_data[filtered_cells]
17
18 print("\nFiltered matrix shape:", filtered_data.shape)
19
20 normalized_data = np.log1p(filtered_data)
21
22 genes_of_interest = ['tp53', 'sod1', 'bcl2a', 'bcl2a', 'bcl2a', 'gpx1a', 'ccnb1', 'ccnd1', 'ctnnb1', 'ugt1a', 'esr1']
23
24 normalized_data.index = normalized_data.index.str.lower()
25 genes_of_interest = [g.lower() for g in genes_of_interest]
26
27 summary_results = {}
28 for gene in genes_of_interest:
29     if gene in normalized_data.index:
30         expr = normalized_data.loc[gene]
31         summary = expr.describe()
32         summary_results[gene.upper()] = summary
33         print(f"\n {gene.upper()} expression summary:\n{summary}")
34     else:
35         print(f" {gene.upper()} not found in matrix.")
36
37 pd.DataFrame(summary_results).T.to_csv("gene_expression_summaries.csv")
38
39
40
```

2.3. Gene Expression Profiling

Genes identified during the 1st objective (including *tp53*, *sod1*, *gpx1a*, *ccnb1*, and *bcl2a*) were extracted from the filtered expression matrix.

Descriptive statistics such as mean, median, and maximum expression levels were calculated for these genes.

2.4. Integration with Network Data

Expression profiles were visually overlaid on the Cytoscape network using expression-based node colouring. This confirmed that hub proteins were not only central but also transcriptionally active in the zebrafish brain at the tested developmental stage, stating their biological relevance.

Colour Pattern

Expression Pattern

Bright Red

Highly expressed in brain baseline

Orange to Yellow

Moderately expresses in brain baseline

Blue

Low or no expression

3.1. Ligand and Protein Structure Preparation

Chlorpyrifos 2D and 3D structures were retrieved from PubChem (CID: 2730) and converted to docking-ready PDB format using Open Babel.

Receptor proteins including tp53, sod1, cyp1a, gpx1a, and bdnf were obtained from the Protein Data Bank (PDB).



Figure: Structure of bdnf

3.2. Docking Execution with CB-Dock2

Performed docking using CB-Dock2, a web-based blind docking platform.

Receptors were uploaded individually and docked with Chlorpyrifos. The tool auto-identified potential binding cavities and used AutoDock Vina to evaluate binding affinity.

Five top-ranked cavities were reported per protein, and the best-scoring conformation was selected.

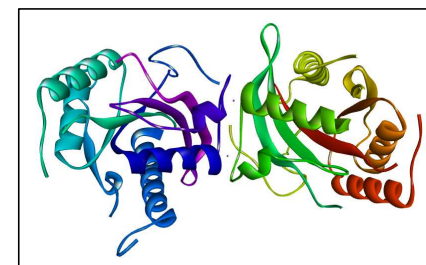


Figure: Structure of tp53

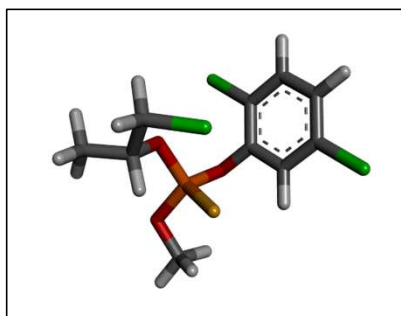


Figure: Stick model of Ligand (Chlorpyrifos)

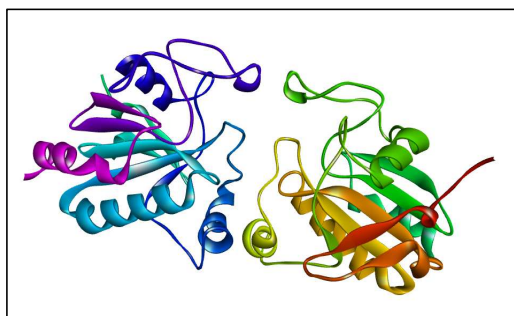


Figure: Structure of gpx1a

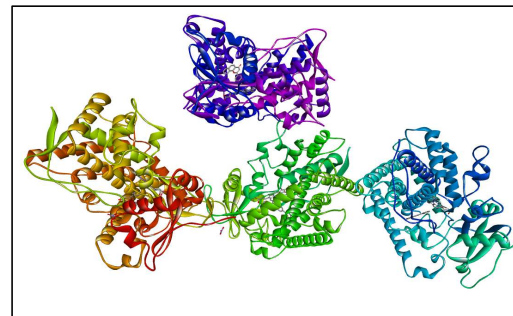


Figure: Structure of cyp1a

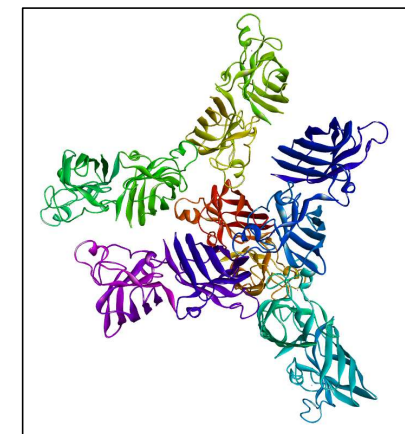
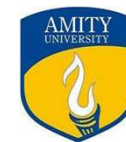


Figure: Structure of sod1





4.1. Feature Compilation

Compiled a dataset with multiple chemicals.
Features used: Molecular Weight (MW), logP, LD₅₀, and number of pathway hits and known toxicity label.

4.2. Normalisation and scaling

All numeric features were normalized using min-max scaling.

$$\text{Scaled Value} = \frac{(\text{Value} - \text{Min Value})}{(\text{Max Value} - \text{Min Value})}$$

4.3. Model Construction in Excel

Implemented a weighted linear combination model.
Model optimization using Solver function.
Finally performance evaluation.

$$\text{Predicted label} = \begin{cases} 0 & , \text{if } \hat{y} < 0.6 \\ 1 & , \text{if } 0.6 \leq \hat{y} \leq 0.8 \\ 2 & , \text{if } \hat{y} \geq 0.8 \end{cases}$$

Class	Sigmoid Range	Toxicity Mapping
0	0.0 to 0.6	Clearly low
1	Between 0.6 and 0.8	Mid/Moderate range
2	0.8 and above	Clearly high

A linear combination of scaled features was calculated and passed through the sigmoid activation function to generate predicted toxicity probabilities.

- *Linear combo (Z) =*

$$\sum (W_{\text{values}} \times \text{Scaled}_{\text{values}}) + \text{Intercept}$$

- *Predicted (Sigmoid) function ($\hat{y}$) = $\frac{1}{1+e^{-Z}}$*
- *Error = $(\hat{y} - y)^2$*

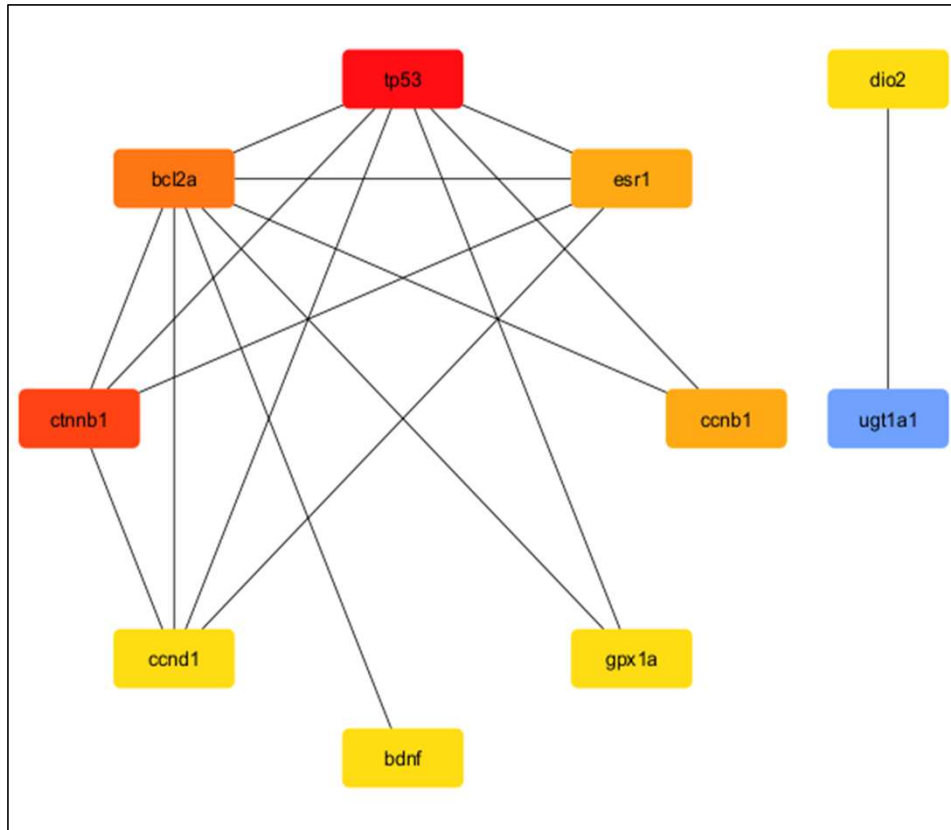
Where, $\hat{y}$ = predicted value from sigmoid
y = true encoded toxicity label (0,1 or 2)

❖ *A confusion matrix (3 X 3) was prepared to calculate accuracy*

$$\text{Accuracy} = \frac{\text{Sum of diagonal values}}{\text{Total number of rows}}$$

RESULTS AND DISCUSSION

Objective 1: Network Biology and Pathway Disruption



Colour Pattern	Expression Pattern
Bright Red	Highly expressed in brain baseline
Orange to Yellow	Moderately expresses in brain baseline
Blue	Low or no expression

Cluster 1- Stress Response & Neurodevelopment Pathways

Pathway or Function	Genes	Toxicological Insight
p53 Signalling / Apoptosis	tp53, bcl2a, ccnb1	Indicates DNA damage, cell cycle arrest, and programmed cell death typical under oxidative or genotoxic stress (e.g., AgNPs, CPF).
Neurotrophins & BDNF Pathway	bdnf	CPF is a neurotoxicant, downregulation of BDNF can impair neuronal survival and function.
Developmental Signalling	ctnnb1, esr1	Suggests disruption of embryonic development, likely affecting reproduction and growth.
Oxidative Stress Regulation	gpx1a	Detoxification of reactive oxygen species (ROS) induced by nanoparticles.

Cluster 2- Detoxification

Pathway or Function	Genes	Toxicological Insight
Biphenyl/Drug Metabolism	ugt1a1	Involved in glucuronidation, critical for conjugating CPF metabolites and xenobiotic clearance.
Hyperbilirubinemia Links (HP Terms)	ugt1a1	Enzyme downregulation or inhibition may lead to toxic buildup, liver stress.

Objective 2: RNA-Seq Expression Validation

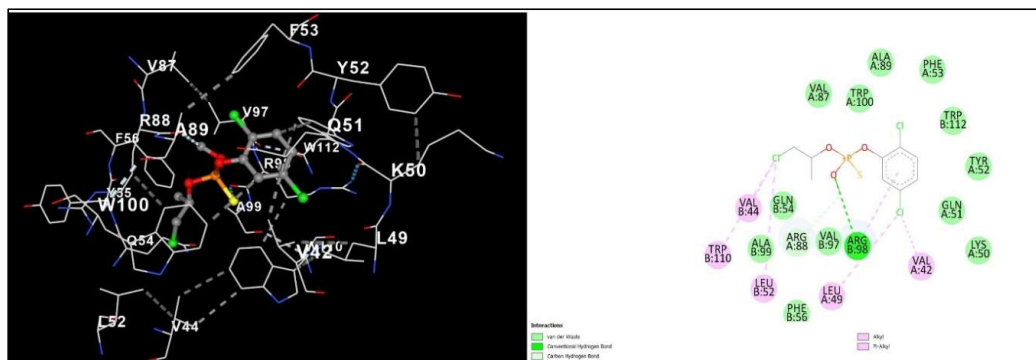
Single-cell RNA-seq analysis confirmed expression of several target genes in zebrafish brain cells at 6.5 dpf and following pattern of hub genes was obtained.

Gene	Mean	Deviation	Min	25%	50%	75%	Max
tp53	0.118885	0.490413	0.000000	0.000000	0.000000	0.000000	3.750709
bcl2a	0.066609	0.389531	0.000000	0.000000	0.000000	0.000000	3.436410
bdnf	0.771309	1.297719	0.000000	0.000000	0.000000	2.070049	4.714498
gpx1a	0.104215	0.468847	0.000000	0.000000	0.000000	0.000000	4.136792
ctnnb1	1.411514	1.457645	0.000000	0.000000	1.518432	2.707117	4.992753
ccnb1	0.056057	0.354046	0.000000	0.000000	0.000000	0.000000	3.610140
ccnd1	0.253669	0.783138	0.000000	0.000000	0.000000	0.000000	4.469349

Gene	Expression	Toxic Insight
tp53	Exceptionally low in most cells	Low basal DNA damage/apoptosis activity; may be activated upon exposure
bcl2a	Extremely low	Consistent with minimal apoptosis at baseline
bdnf	Moderate to high (mean 0.77, max 4.7)	Active neuroprotective signal, CPF may downregulate this
gpx1a	Low	Indicates low oxidative stress under control; may increase after AgNP exposure
ctnnb1	Strong (mean 1.4, median ~1.5)	Active Wnt signalling/developmental regulation in brain
ccnb1	Extremely low	Low proliferation in mature neurons expected
ccnd1	Low-moderate	Some cell cycle activity, possibly in progenitors or glial cells
esr1, ugt1a1	Not detected	Could be low expression in brain or filtered out in QC

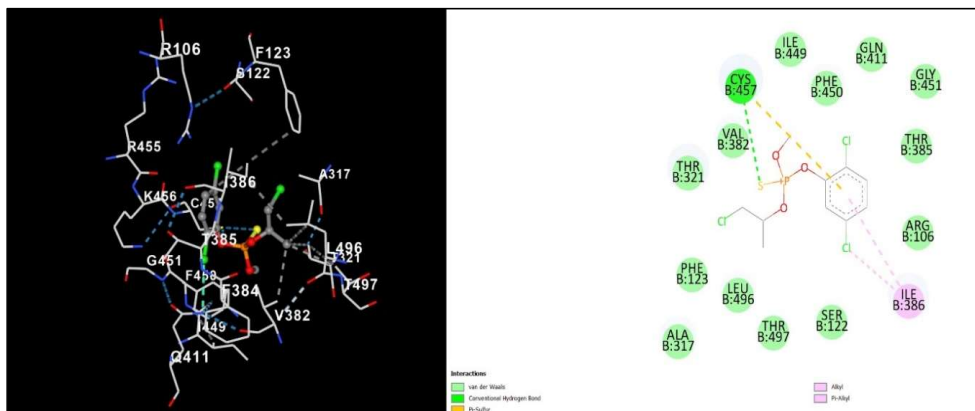
Objective 3: Molecular Docking Insights

Ligand	Receptor	Score (kcal/mol)	Cavity volume (Å ³)	Center (x, y, z)	Docking size (x, y, z)	Key Residues Involved
Chlorpyrifos	bdnf	-5.7	1200	0, 36, 2	19, 19, 19	ARG:B98, ALA:B99, TRP:B110, LEU:A49
Chlorpyrifos	cyp1a	-6.0	2260	-26, 84, -5	19, 19, 19	CYS:B457, PHE:B450, ILE:B386, GLN:A78
Chlorpyrifos	gpx1a	-4.2	577	-15, 19, 28	19, 19, 19	GLU:A114, GLN:A78, ASN:A77, THR:A149
Chlorpyrifos	sod1	-5.9	1862	75, 8, -8	29, 19, 19	VAL:H148, ASN:H53, CYS:H146, LYS:G9
Chlorpyrifos	tp53	-5.6	231	-47, 58, -6	19, 19, 19	ARG:A238, THR:A3, ILE:A224, GLN:A240



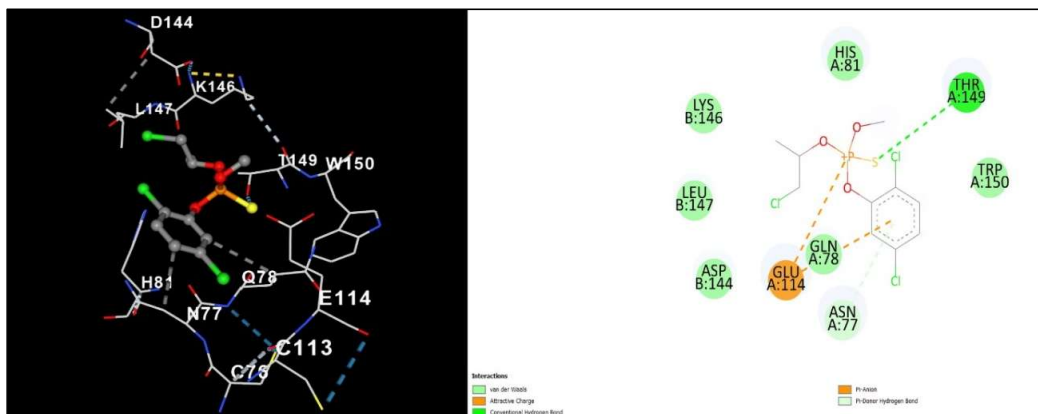
Docking of bdnf protein with chlorpyrifos (a) 3D Conformation
(b) 2D Conformation

BDNF displayed a binding score of -5.7 kcal/mol. Interaction mapping revealed a strong hydrogen bond with ARG:B98, while hydrophobic interactions were observed with TRP:B110, ALA:B99, and LEU:A49. The binding cavity encompassed multiple π -alkyl interactions, indicating a stable insertion of the chlorinated aromatic moiety of the ligand. Given BDNF's critical role in neuronal development and synaptic plasticity, Chlorpyrifos binding may impair neurotrophic signalling, potentially contributing to developmental neurotoxicity.



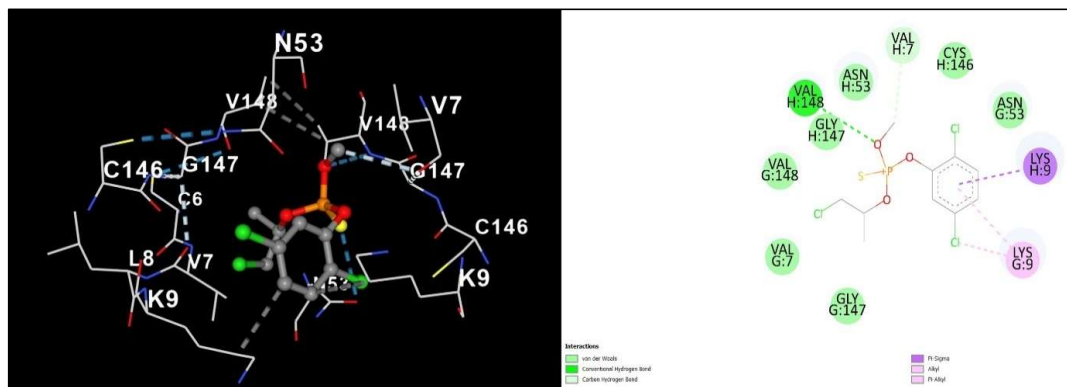
Docking of cyp1a with chlorpyrifos (a) 3D Conformation
(b) 2D Conformation

Chlorpyrifos exhibited its highest binding affinity (-6.0 kcal/mol) with CYP1A. The docking pose demonstrated strong stabilization through a hydrogen bond with CYS:B457, along with π -sulfur interactions involving GLN:A78 and PHE:B450. Hydrophobic contacts with ILE:B386 and VAL:B382 suggest occupation of the substrate-binding region. Since CYP1A is a key player in xenobiotic metabolism, especially phase I biotransformation, the binding of Chlorpyrifos may competitively inhibit its enzymatic function, leading to the accumulation of reactive or unmetabolized intermediates.



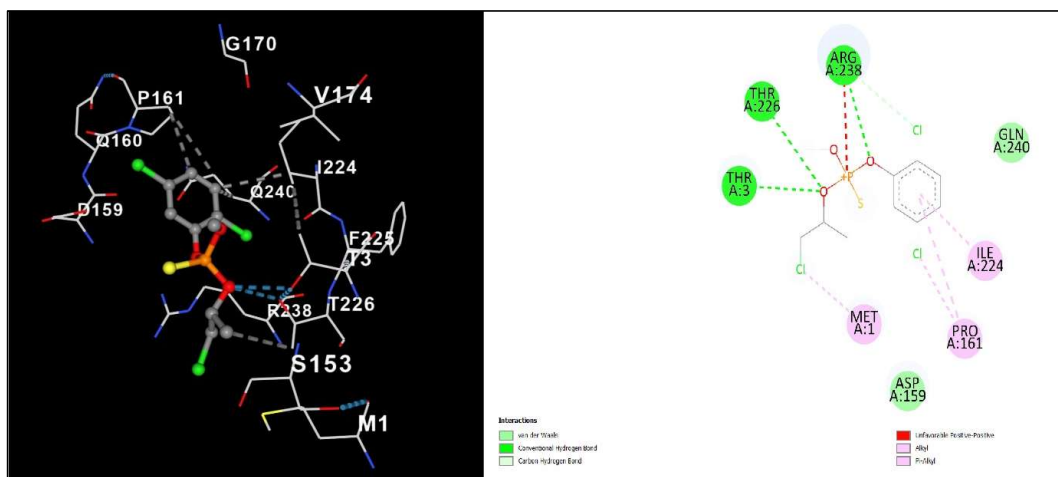
Docking of gp1a with chlorpyrifos (a) 3D Conformation
(b) 2D Conformation

GPX1A, an antioxidant enzyme involved in neutralizing hydrogen peroxide, showed the lowest affinity (-4.2 kcal/mol) among the targets. Hydrogen bonding was observed with THR:A149 and ASN:A77, and π -anion and π -donor interactions with GLU:A114 and GLN:A78, respectively. The relatively modest score and limited contacts imply a lower likelihood of direct GPX1A inhibition, although cumulative effects with other oxidative enzymes may still contribute to redox imbalance.



Docking of sod1 with chlorpyrifos (a) 3D Conformation
(b) 2D Conformation

The Chlorpyrifos–SOD1 complex showed a -5.9 kcal/mol docking score. The pesticide engaged in conventional hydrogen bonding with VAL:H148, ASN:H53, and CYS:H146, located near the enzyme's active site. Pi-alkyl interactions with LYS:G9 contributed additional stabilization of the ligand within the binding pocket. The presence of a dense network of van der Waals contacts indicates strong geometric complementarity between the ligand and the protein surface. Considering SOD1's critical role in neutralizing reactive oxygen species (ROS), such interactions may impair its enzymatic activity, potentially leading to elevated oxidative stress in zebrafish tissues following exposure.



Docking of tp53 with chlorpyrifos (a) 3D Conformation
(b) 2D Conformation

Docking of Chlorpyrifos with TP53 revealed a binding score of -5.6 kcal/mol. The ligand established several hydrogen bonds with critical residues ARG:A238, THR:A226, and THR:A3—suggesting a stable interaction within the DNA-binding domain of TP53. Hydrophobic interactions with ILE:A224, MET:A1, and PRO:A161 may further contribute to the overall binding affinity and complex stability. Notably, an unfavourable positive-positive interaction was noted between ARG residues, potentially destabilizing the complex in specific conformations. These results suggest that Chlorpyrifos may interfere with the DNA-binding activity or structural integrity of TP53, potentially impairing cell cycle regulation or apoptosis.

Objective 4: Machine Learning Toxicity Prediction

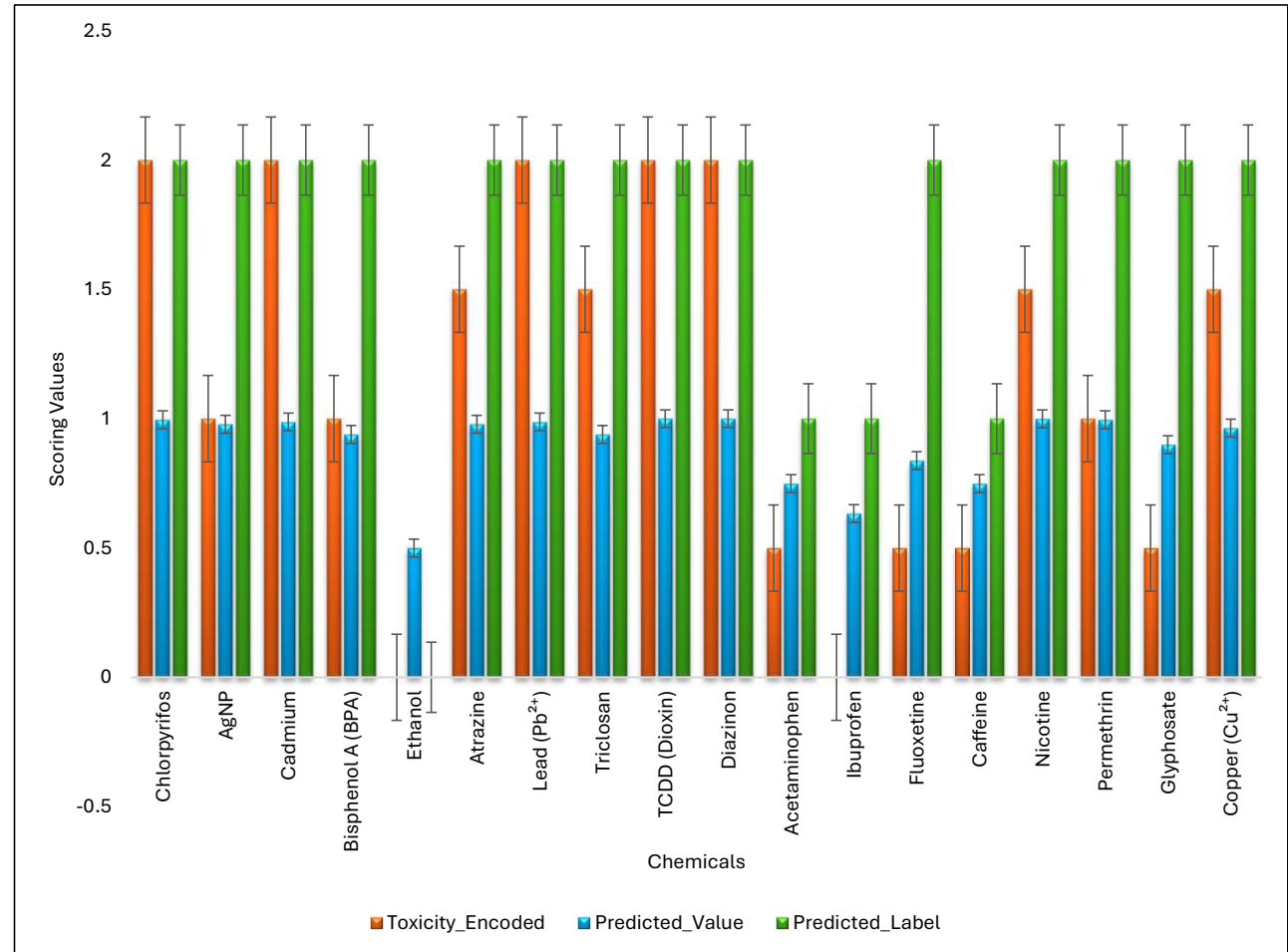
A confusion matrix (3 X 3) was created as:

	Predicted 0	Predicted 1	Predicted 2
True 0	1	1	0
True 1	0	2	5
True 2	0	0	9

$$Accuracy = \frac{1+2+9}{18} = \frac{12}{18} = 0.6667 = 66.7\%$$

The prediction model that showed 66.7% accuracy is quite acceptable and meaningful in conventional framework and goals of this study. The model was developed with minimum features representing reductionist yet interpretable approach suitable for early stage screening. The dataset contained diverse chemical classes, this heterogeneity increases complexity but is highly specific due to narrow values.

Model Validation: Actual vs. Predicted Toxicity



Chemical	Actual Label	Predicted Label	Predicted Value	Result
TCDD	2	2	1.000	Correct prediction. Strong alignment with known high toxicity
Diazinon	2	2	1.000	Correct prediction. Matches established toxicity classification
Nicotine	1.5	2	0.999	High prediction. Matches intermediate-high toxicity and pathway complexity
Chlorpyrifos	2	2	0.996	Correct. Model effectively captured high-risk pesticide
Permethrin	1	2	0.996	Overpredicted. High pathway involvement likely pushed prediction toward high toxicity, despite moderate LD ₅₀ .
Cadmium	2	2	0.988	Correct prediction. Heavy metal toxicity well reflected
Lead (Pb ²⁺)	2	2	0.988	Correct. Model successfully predicted neurotoxic effects
AgNP	1	2	0.979	Slight overprediction. May reflect unknown chronic mechanisms
Atrazine	1.5	2	0.979	Overpredicted. Possibly due to moderate pathway activity
Copper (Cu ²⁺)	1.5	2	0.964	Overpredicted. May be justified by neurotoxic pathway engagement
Triclosan	1.5	2	0.939	Overpredicted. Pathway hits may have inflated toxicity estimate
Bisphenol A	1	2	0.939	Overpredicted. Moderate risk assigned higher due to sensitivity to pathway data
Glyphosate	0.5	2	0.899	Overpredicted. High predicted score despite moderate real-world classification
Fluoxetine	0.5	2	0.838	Overpredicted. Borderline value just above threshold
Acetaminophen	0.5	1	0.749	Accurate. Borderline moderate classification
Caffeine	0.5	1	0.749	Accurate. Moderate-low toxicity represented well
Ibuprofen	0	1	0.633	Slightly overpredicted, but still near threshold
Ethanol	0	0	0.500	Correct. Well within low toxicity prediction range

CONCLUSION

- ❖ This study established a comprehensive computational strategy to assess the long-term toxicity of chlorpyrifos (CPF) and silver nanoparticles (AgNPs) in zebrafish. It began by identifying key toxicant-associated genes from CTDBase and mapping their interactions through protein-protein network analysis using STRING and Cytoscape. Functional modules and hub genes like TP53, SOD1, and BDNF were identified and enriched for biological relevance.
- ❖ Integration of single-cell RNA-seq data allowed validation of these genes' expression in specific brain cell types, revealing their potential involvement in stress and neurotoxicity. Molecular docking simulations further supported the hypothesis by showing strong binding affinities of CPF to zebrafish detoxification and oxidative stress-regulating proteins.
- ❖ A simplified predictive machine learning model was built in MS Excel using physicochemical and pathway-based features. This model achieved a 66.7% prediction accuracy, showing reasonable performance given its interpretable structure and limited feature set. Among all features, pathway hits emerged as the most influential factor, highlighting the value of omics-informed data in toxicity scoring.
- ❖ Each layer of analysis—network, transcriptomic, docking, and modeling, contributed independently and synergistically to build a cohesive risk assessment model. The project reaffirmed zebrafish as a powerful systems-level model for environmental toxicology.
- ❖ Ultimately, the study presents an ethical, scalable, and mechanistically grounded approach that supports next-generation chemical risk evaluation in the absence of exhaustive experimental testing.

FUTUTRE PERSPECTIVES

- ❖ This study sets the stage for future advances in computational toxicology, especially for nanomaterials and complex chemical mixtures.
- ❖ Future work can include incorporating molecular dynamics to better simulate AgNP interactions at atomic resolution.
- ❖ Expanding transcriptomic data across developmental stages and integrating proteomics/metabolomics will enhance mechanistic insights.
- ❖ The current MS Excel-based model can be upgraded using advanced ML platforms like Orange ML or TensorFlow to improve accuracy and scalability for larger datasets.
- ❖ With a larger training dataset from sources like ToxCast and PubChem, model generalizability will improve significantly.
- ❖ Developing this pipeline into a user-friendly decision-support tool can assist regulatory screening and reduce animal testing.
- ❖ Additionally, linking in vivo zebrafish endpoints with predicted disruptions can support adverse outcome pathway (AOP) development and regulatory acceptance. Overall, this approach holds strong promise for sustainable, ethical, and scalable toxicology research.

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